

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

26th June 2019

Subject CB1 – Business Finance

Time allowed: 3 Hours 15 Minutes (10.15 – 13.30 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

- 1. Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
- 2. Mark allocations are shown in brackets.*
- 3. Attempt all questions, beginning your answer to each question on a separate sheet. However, answers to objective type questions could be written on the same sheet.*
- 4. Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

- Q. 1)** The distinction between an American Option and European Option is that
- A.** Under American Option right can be exercised at any time while European right can be exercised on a specific date
 - B.** Under American Option right can be exercised on a specific date while European right can be exercised at any time
 - C.** Under both the options rights can be exercised only at a specific time
 - D.** Under both the options rights can be exercised on any data [2]
- Q. 2)** Peer to Peer Lending is a type of
- A.** Shadow Banking
 - B.** Project Financing
 - C.** Non-Recourse Financing
 - D.** Crowdfunding [2]
- Q. 3)** Recognition of a lease as a finance lease as opposed to an operating lease by the lessee will most likely result in a higher:
- A.** Debt-to-assets ratio.
 - B.** Return on assets.
 - C.** Current ratio.
 - D.** Acid Test Ratio [2]
- Q. 4)** Statement 1: Apart from normal dividends, cumulative preference shares get an additional dividend if the company's profits exceed a pre-specified level.
- Statement 2: From the investor's perspective, callable common shares are more risky than puttable common shares.
- Which of the following is most likely?
- A.** Only Statement 1 is incorrect.
 - B.** Only Statement 2 is incorrect.
 - C.** Both statements are correct.
 - D.** Both Statements are incorrect [2]
- Q. 5)** How might the interest of the company's management be NOT aligned with those of shareholders
- A.** Linking the management remuneration to the company's earning
 - B.** Linking the management remuneration to the company's share price
 - C.** Linking the management bonus to the company's share price
 - D.** Linking the management bonus to the company's Credit Rating [2]
- Q. 6)** A health insurance company allows an inward and outward reinsurance in its business. Which amongst the following in revenue account will be directly impacted.
- A.** Net Earned Premium
 - B.** Net Incurred Claims
 - C.** Share Capital
 - D.** Option (A) and (B) [2]

- Q. 7)** Any company's management wish to receive what sort of opinion from auditors on its accounts?
- A.** An unqualified opinion
 - B.** A qualified opinion
 - C.** Adverse opinion
 - D.** None of the above
- [2]**
- Q. 8)** Which of the option is correct using as an approach to forecasting?
- A.** Time series analysis.
 - B.** Seeking an expert's opinion.
 - C.** Delphi technique.
 - D.** All of the above
- [2]**
- Q. 9)** The cash-flow position can be improved by?
- A.** Deferring trade receivables payments
 - B.** Fast payment of trade payables
 - C.** Disposing of an asset
 - D.** All of the above
- [2]**
- Q. 10)** A company has INR 500,000 line of credit at 10.0% pa with a 1.0% pa commitment fee on the full amount available. The company draws down INR 100,000 for 6 months. The annual financing cost of this arrangement is:
- A.** 7.5%
 - B.** 15.0%
 - C.** 12.5%
 - D.** 10.0%
- [2]**
- Q. 11)** Explain the meaning of the term 'stakeholder', providing examples of stakeholders in your answer.
- Outline the benefits of corporate social responsibility (CSR) for companies.
- [5]**
- Q. 12)** A gaming company based out of Hong Kong is considering building and operating a game-based theme park in India. Explain the process of conducting a simulation for financial evaluation of this project.
- [5]**
- Q. 13)** Describe the disadvantages of establishing a business as a traditional partnership rather than limited liability partnership for professional services.
- [5]**
- Q. 14)** How does investor perception contribute to set the dividend policy? Highlight the information you would require for setting dividend strategy during recession in the industry the company is operating in and also the possible course of action.
- [5]**
- Q. 15)** The directors of ABC Ltd are considering a five-year project.
- The COO proposes to buy a used machine which has an expected remaining useful life of five years for Rs 3 million.

The sales director proposes taking out a five-year lease on a factory building and using a recruitment firm to provide manpower for the five-year period. The director estimates that it would be necessary to invest a further Rs 1 million at the start of the first year in addition to buying the machine: a returnable deposit of Rs 0.1 million on the factory, Rs 0.2 million for the first year's lease, Rs 0.4 million for the opening inventory of raw materials and Rs 0.3 million for the initial payment for man power.

The sales director has drafted the following budgeted annual income statement for the project:

	Rs million
Revenue	2.5
Raw materials	-0.4
Factory lease	-0.2
Labour	-0.3
Depreciation of machine	-0.8
Other running costs	-0.5
Profit	0.4

It is anticipated that materials, recurring lease payments and payment to employees will be paid for annually at the start of each year. Other running costs comprise the cost of electricity and other operating costs and they will be paid for annually at the end of each year.

Revenues will be received at the end of the year in which they are earned. At the conclusion of the project, the sales director anticipates that it will cost Rs 0.3 million to dismantle the equipment and at that time it will be possible to reclaim 100% of the deposit paid to the factory owner.

The sales director has shown this proposal to a potential lender, who has agreed to lend Rs 5 million at an interest rate of 7% p.a.

The sales director proposes that the project should be evaluated at a required rate of return of 7% p.a. because it will be financed by means of a self-contained loan package upon which that rate will be charged. The finance director believes that the project should be evaluated at a much higher rate than 7% p.a. but cannot state the specific rate that should be charged without conducting a further analysis.

The sales director has ignored tax while preparing the budgeted income statement.

- i) Calculate the net present value of the project using a discount rate of 7% p.a. and ignoring tax. (7)
- ii) Explain why the required rate of return might be higher than the 7% rate charged by the bank on the funding for the project. (6)
- iii) Calculate the net present value assuming the running cost increases by 10% and the discount rate is 9%. (5)
- iv) Explain how allowing for tax may affect the cost of capital. (2)

[20]

- Q. 16)** i) What do you understand from Prudence in accounting concepts? (2)
- ii) Explain, how and in which sections of account a general insurance company's actuary should utilize this concept? (3)
[5]
- Q. 17)** Compare statistical approaches to forecasting with intuitive approaches such as Delphi techniques. [5]
- Q. 18)** i) Define the term gross profit, operating profit and profit after tax? (3)
- ii) What gross profit indicates and in what circumstances a company can have gross profit but no operating profit. (2)
[5]
- Q. 19)** If you want to measure any business efficiency, what ratios you will analysis and why? [6]
- Q. 20)** As an account officer, you have been requested to assist ABC Ltd., a company, to prepare its annual accounts. The following information has been provided from the company's Book-keeping records:

Balances as on 31 March 2019

Administrative staff wages: 3,000
 Cost of Marketing: 55,000
 Bank overdraft: 18,000
 Buildings – depreciation: 40,000
 Buildings – valuation: 400,000
 Cost of inventory consumed: 350,000
 Delivery vehicle running costs: 60,000
 Delivery vehicles – cost: 325,000
 Delivery vehicles – depreciation: 170,000
 Dividend paid: 45,000
 Factory running costs: 100,000
 Interest: 10,000
 Inventory at 31 March 2019: 30,000
 Land – valuation: 600,000
 Loan (repayable 2024): 145,000
 Machinery – cost: 150,000
 Machinery – depreciation: 80,000
 Manufacturing wages: 175,000
 Retained earnings: 250,000
 Revaluation reserve: 180,000
 Revenue: 1,500,000
 Sales salaries: 120,000
 Share capital 200,000
 Trade payables: 50,000
 Trade receivables: 210,000

In addition to figures shown above, below information is provided:

1. A revaluation exercise was conducted on 1 April 2018. Land was revalued at 800,000 and buildings at 700,000
2. Depreciation has still to be charged as follows:
 - Buildings – 1% of cost or valuation
 - Delivery vehicles – 20% reducing balance
 - Machinery – 10% of cost

Prepare:

- | | |
|---|-------------|
| i) An income statement for the year ended 31 March 2019 | (9) |
| ii) A statement of changes in equity, and | (4) |
| iii) A statement of financial position as on 31 March 2019 | (6) |
| | [19] |
